



User Guide — monitor and manage your Bitcoin miner fleet
from your Android phone, tablet, or TV

Guide for app version 0.73.4 · September 2026
Local-first: the app talks only to your miners on your own network.

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1 · Welcome & key ideas

Hi3 Hashkit turns a spare Android device into a command center for a home or small-site Bitcoin mining fleet. It speaks directly to each miner's own API over your LAN (or Tailscale), so there is no cloud account, no sign-up, and no data leaving your network unless you explicitly opt in to an integration.

- **Local-first & private.** Telemetry, alerts, notes and history are stored on the device. Backups can be encrypted; exports redact wallet addresses.
- **Multi-vendor.** Bitaxe / NerdQAxe (AxeOS & forks), Canaan Avalon (Nano 3 and others), Braiins OS devices, and Antminer-class firmwares (monitoring).
- **Safe by design.** Controls are only enabled where the firmware's API has been verified; everything else stays monitoring-only. Risky actions always confirm first.
- **Demo mode.** Settings → Demo adds clearly-labeled synthetic miners (used for the pictures in this guide) so you can explore every screen without hardware.

2 · Getting started — install & first launch

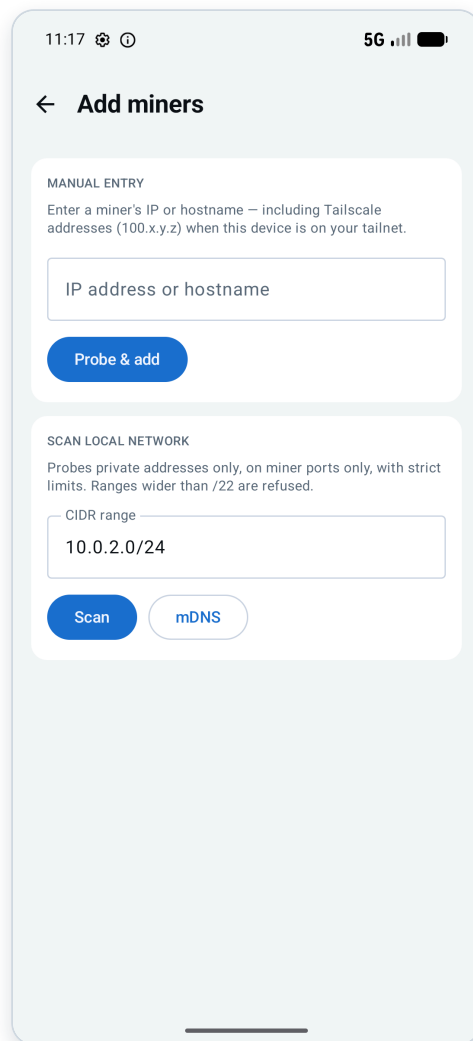
Hashkit is distributed as a signed APK on the project's GitHub releases page (sideload). On your Android device:

- Download the latest `Hi3Hashkit-vX.Y.Z-release.apk` from the releases page and open it. Android will ask to allow installs from your browser or file manager the first time.
- Later updates are much easier — the app updates itself from inside **Settings** → **Update Hashkit** (see section 10).

- On first launch a short onboarding introduces the app; you can turn on **Demo mode** there or in Settings to explore with simulated miners.

Tip: minimum Android 8.0 (API 26). Phones, tablets, and Android TV all work — there's a purpose-built Wall / TV mode for room displays.

3 · Adding your miners



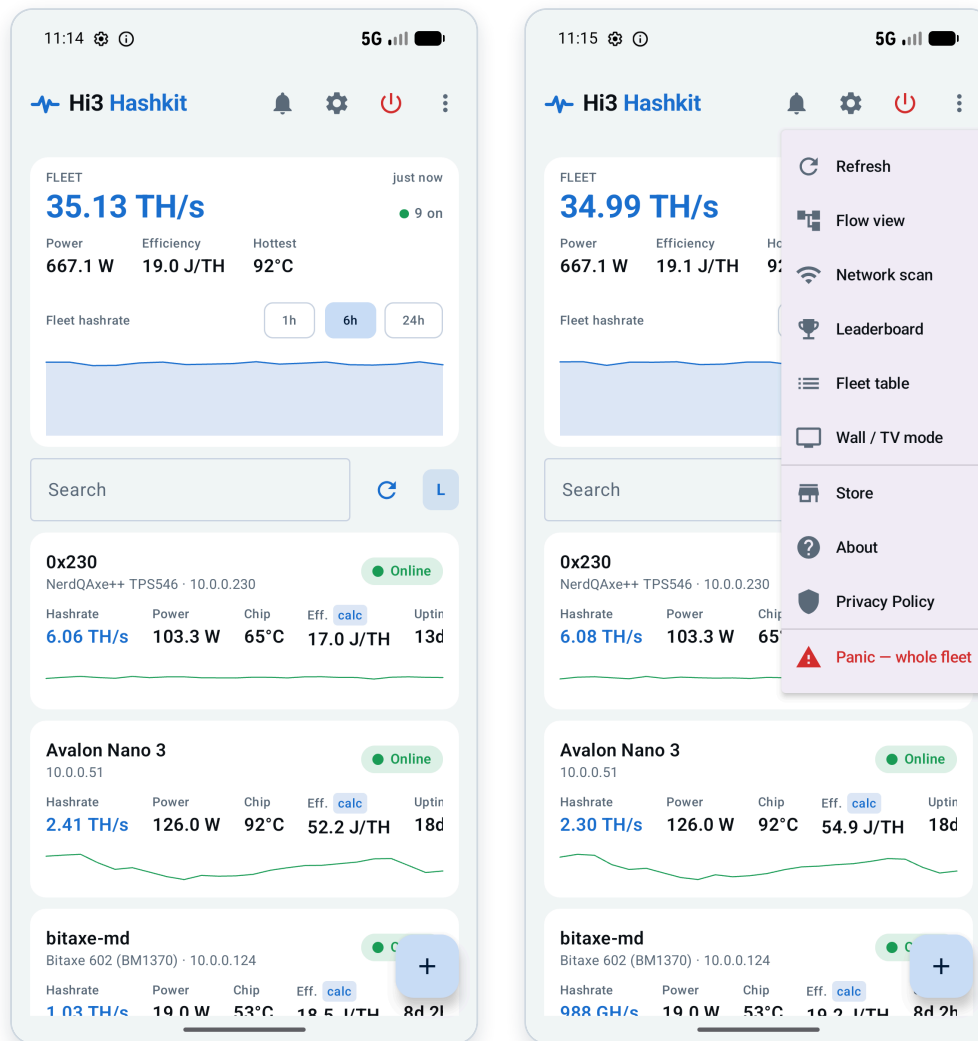
Add miners — manual entry, subnet scan, or mDNS discovery

Tap the + button on the dashboard. You have three ways to add hardware:

- **Manual entry.** Type the miner's IP or hostname — Tailscale addresses (`100.x.y.z`) work when the device is on your tailnet. **Probe & add** detects the firmware type automatically.
- **Scan local network.** Give a CIDR range (e.g. `10.0.0.0/24`) and Hashkit probes private addresses on miner ports only, with strict limits — ranges wider than /22 are refused.
- **mDNS.** Discovers miners that announce themselves (e.g. AxeOS devices) with no typing at all.

Miners are identified by model, ASIC and firmware after the first probe, and each becomes a card on the dashboard.

4 · The Dashboard

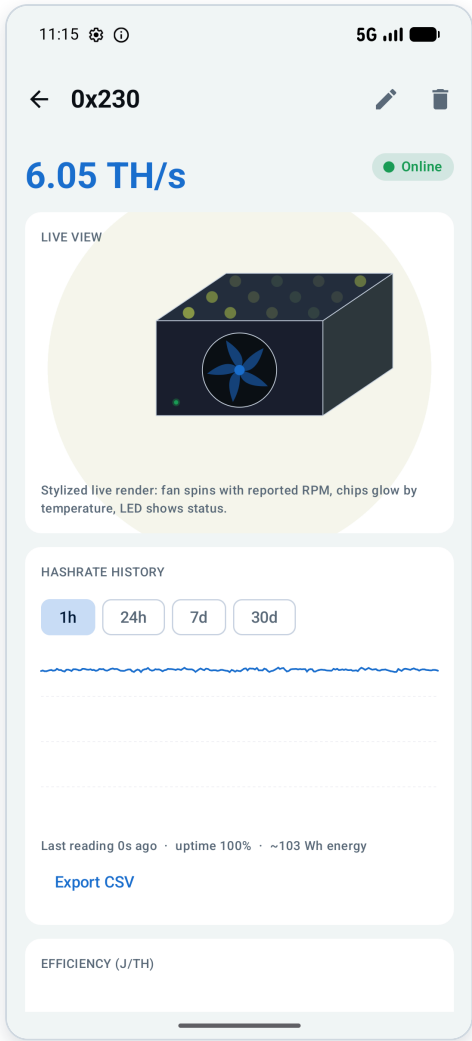


Fleet summary, trend chart and per-miner cards with sparklines

The $\vdots$ menu: fleet views, network scan, panic action

- **Fleet card.** Total hashrate, power, efficiency (J/TH), hottest chip and online count, with a 1h / 6h / 24h fleet trend chart.
- **Miner cards.** One per device: live hashrate, power, chip temperature, efficiency and uptime plus a sparkline. Tap a card for full detail. Green/red dots show online state at a glance.
- **Search & groups.** Filter by name, and use groups (long-press to multi-select) for bulk actions — pause/resume, reboot and more, always with a capability preview so you see exactly which miners support the action.
- **Density.** The **L** button next to search cycles card density (Large / Medium / Small) to fit more miners per screen.
- **Panic — whole fleet.** At the bottom of the menu: one action to stop everything that supports being stopped, for emergencies. It confirms before running.

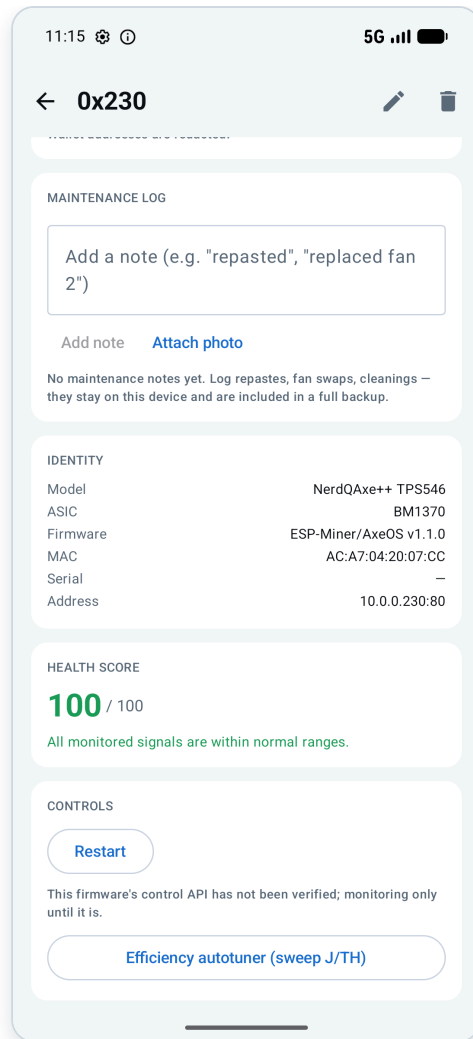
5 · Miner detail — telemetry, health & controls



Live view — the fan spins at real RPM, chips glow by temperature



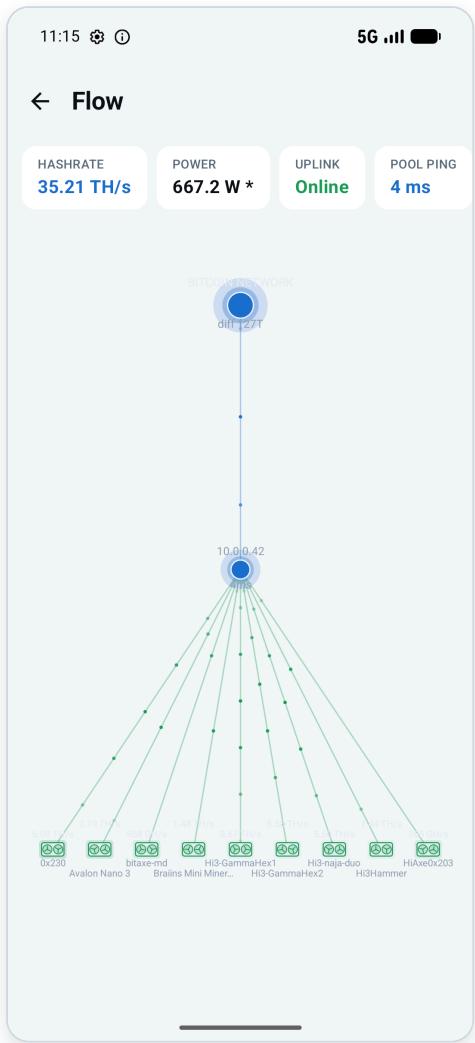
Efficiency history, live telemetry, shares & pool, live logs



Maintenance log, identity, health score and verified controls

- **Live view.** A stylized live render of the unit: fan speed, per-chip temperature glow and status LED update with each poll.
- **Charts.** Hashrate history over 1h / 24h / 7d / 30d and an efficiency (J/TH) chart with min / now / max. Any chart's window exports to CSV.
- **Live telemetry.** Power, efficiency, chip & VR temperatures, frequency, core voltage, fan RPMs, uptime — everything the firmware reports.
- **Shares & pool.** Accepted/rejected shares, best difficulty this session and all-time, and the pool the miner is pointed at.
- **Live logs.** Streams the firmware's own log over WebSocket (AxeOS family). Wallet addresses are redacted automatically.
- **Maintenance log.** Notes with photos per miner — "repasted", "replaced fan 2" — kept on-device and included in backups.
- **Health score.** A 0–100 roll-up of temperature, hashrate vs. expected, rejects, fan health and restarts, with the reasons listed when it drops.
- **Controls.** Only what's verified for that firmware: restart, pause/resume, tuning presets, or the efficiency autotuner (sweeps frequency/voltage to find the best J/TH). Unverified firmwares stay monitoring-only — the app tells you why.

6 · Fleet views — Flow, Table, Leaderboard, Wall/TV

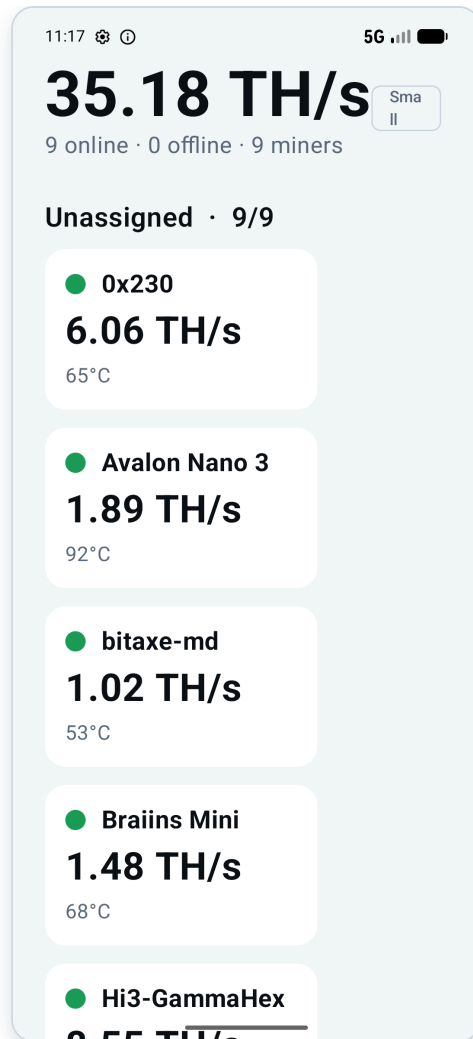


Flow — shares stream from your miners through the pool to the network

The 'Fleet table' view shows a table with columns: Name, IP, Model, and Hashrate. A filter input is available at the top. The table contains the following data:

Name ▲	IP	Model	Hashrate
0x230	10.0.0.230	NerdQAxe++	6.08 TH
Avalon Nano 3	10.0.0.51	Avalon Nano 3	2.19 TH
bitaxe-md	10.0.0.124	Bitaxe 602	958 GH
Brains Mini Miner	10.0.0.168	Brains Mini Miner	1.48 TH
Hi3-GammaHex1	10.0.0.47	Bitaxe 1300	8.67 TH
Hi3-GammaHex2	10.0.0.107	Bitaxe 1300	8.53 TH
Hi3-naja-duo	10.0.0.208	Bitaxe 1201	5.50 TH
Hi3Hammer	10.0.0.148	BC01	1.44 TH
HiAxe0x203	10.0.0.203	Bitaxe 204	365 GH

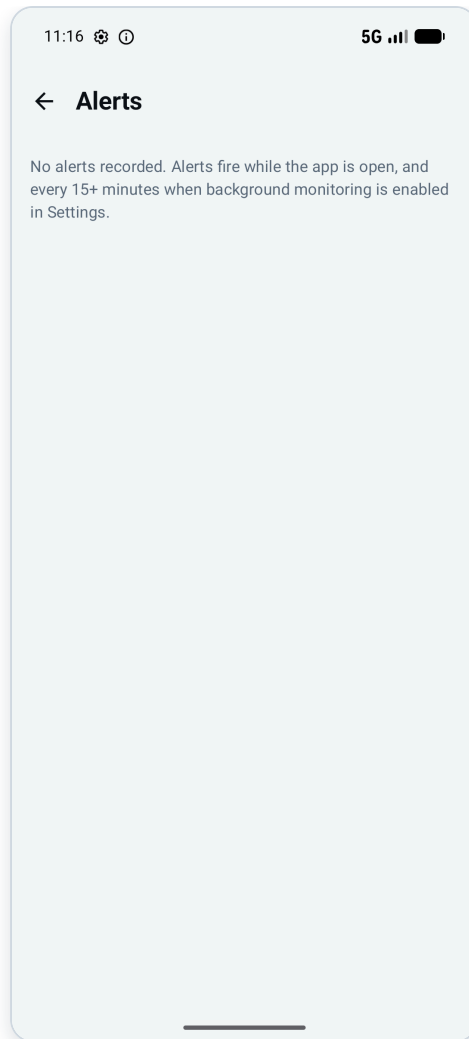
Fleet table — sortable, filterable, built for big fleets



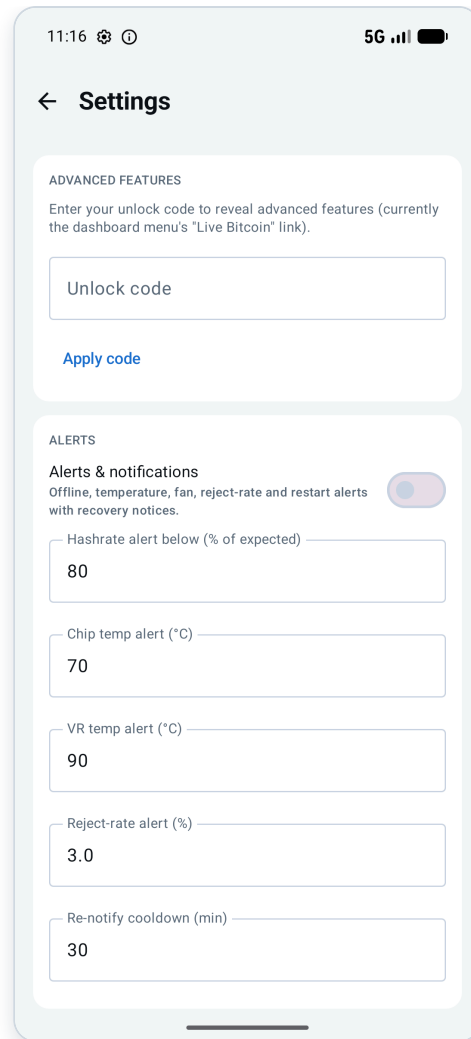
Wall / TV mode — room-readable tiles with a size selector

- **Flow view.** An animated live diagram: each miner's share particles flow toward your pool/proxy and on to the Bitcoin network, sized by hashrate. The header shows fleet hashrate, power, uplink status and pool ping.
- **Fleet table.** A dense sortable table (name, IP, model, hashrate, temps...) with a filter box — the fastest way to find one machine in a big fleet.
- **Leaderboard.** Ranks your miners by best difficulty — the fun view for solo miners chasing a block.
- **Wall / TV mode.** Giant tiles grouped by your miner groups, readable from across a room, with Small / Medium / Large sizing on-screen. Works great on Android TV; the app appears in the TV launcher.

7 · Alerts & background monitoring



The Alerts screen collects every event with quick actions



Alert thresholds are fully adjustable, globally and per miner

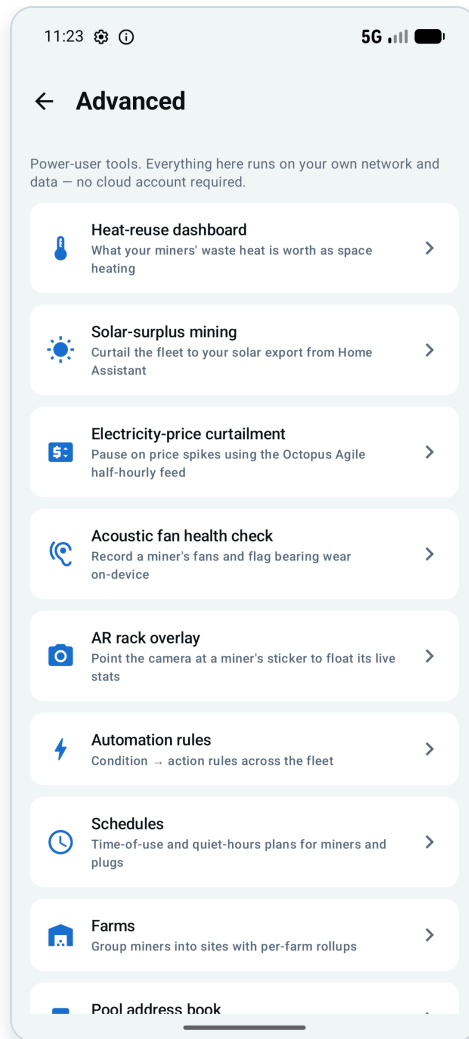
- **What fires.** Offline / back-online, hashrate below a % of expected, chip & VR over-temperature, fan failure or stall, reject-rate, unexpected restarts, statistical anomalies (gradual drift over hours), and firmware-update-available (opt-in).
- **Where.** In-app on the Alerts screen (bell icon) and as Android notifications with per-category channels — so you can silence, say, firmware notices but keep offline alerts loud. Offline and reboot notifications carry quick actions (Acknowledge / Reboot).
- **Per-miner overrides.** Any miner can have its own thresholds, or be muted entirely (good for a bench unit).
- **Background monitoring.** With the toggle on in Settings, a WorkManager job re-checks the fleet every 15+ minutes while the app is closed. A re-notify cooldown stops alert spam.
- **Safety cutoff.** If a miner has a smart plug configured with an over-temp cutoff, Hashkit switches the plug off when the chip temperature crosses your limit — turning it back on is always manual.

8 · Settings that matter

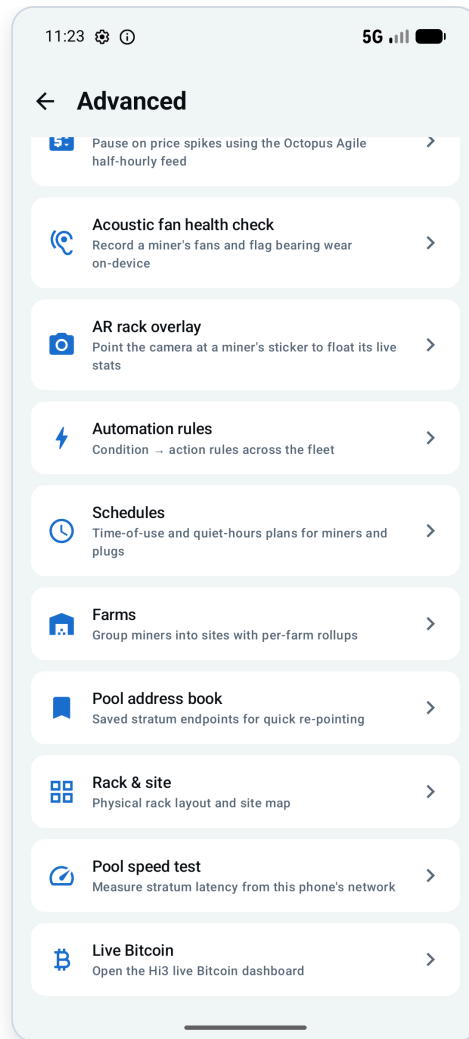
- **Alerts.** Master toggle plus the thresholds shown above.
- **Polling.** Refresh interval (and per-site overrides), extra scan subnets for discovery.
- **Appearance.** System / Dark / Light theme; the accent follows the Hi3 brand. Card density is on the dashboard itself.
- **App lock.** Biometric lock with a 60-second background grace period.
- **Integrations (all opt-in, all off by default).**
 - **Hi3 Pool** — enter your payout address and the dashboard compares miner-side vs. pool-side hashrate per worker.
 - **Hi3 MMP** — read-only fleet summary from your MMP server with an API key (stored encrypted via Android Keystore). The card shows *Local* vs. *MMP fleet-wide* hashrate on one line.
 - **BTC price** — a single HTTPS GET to estimate revenue/profit, or enter a price manually.
 - **Firmware update check** — one GitHub query per 6 h for the latest AxeOS release; shows a banner and a daily per-miner alert when a Bitaxe is behind. It never flashes anything (see section 7).
- **Demo mode.** Adds clearly-labeled synthetic miners; they never mix with real totals when off.

9 · The Advanced hub — power-user tools

Advanced features are unlocked with a code in **Settings** → **Advanced features** → **Apply code**. Once unlocked, an **Advanced** entry appears in the dashboard's : menu and opens the hub below. Everything here runs on your own network and data — no cloud account required.



The Advanced hub (top) — energy, acoustics and AR tools



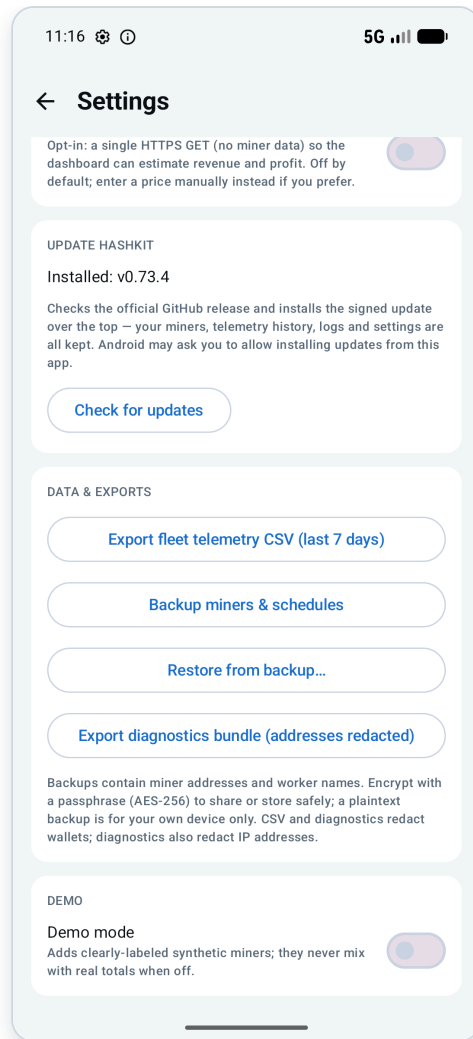
The Advanced hub (continued) — fleet organisation and utilities

Feature	What it does
Heat-reuse dashboard	Values your fleet's waste heat as space heating: live BTU/hr, thermal kWh/day, and what that heat is worth in cash compared to running a resistive heater — and compared to a heat pump — using your electricity price.
Solar-surplus mining	Reads a surplus-power sensor from your local Home Assistant (the access token is stored encrypted; only private-network hosts are allowed) and recommends — or automatically applies — fleet curtailment so the miners soak up excess solar. A hysteresis band stops them flapping on/off as clouds pass.
Electricity-price curtailment	Pulls the free public Octopus Agile (UK) half-hourly electricity price feed and recommends or applies curtailment during price spikes, resuming when prices fall. Shares its curtailment engine with solar-surplus mining.
Acoustic fan health check	Records a few seconds of a miner's fan noise and runs an on-device FFT to flag blade imbalance or bearing wear before a fan fails. Audio is analysed on the phone and never recorded to a file or sent anywhere. Also reachable from a miner's detail screen.

AR rack overlay	Continuous camera QR scan: stick a QR label on each unit, point the camera at the rack, and each miner's live hashrate/temperature floats over the hardware. Tap a bubble to open that miner.
Automation rules	Condition → action rules across the fleet — e.g. "if chip temp > 75 °C for 10 min, pause", "if offline 30 min, cycle the smart plug". Rules run on-device against each poll.
Schedules	Time-of-use and quiet-hours plans for miners and smart plugs — pause overnight, run only in cheap-rate windows, keep the office quiet during calls.
Farms	Group miners into physical sites with per-farm rollups (hashrate, power, health) and per-farm refresh intervals — useful when a remote site sits behind a slower link.
Pool address book	Saved stratum endpoints (URL, port, worker template) so re-pointing one miner or the whole fleet to another pool is a two-tap job instead of retyping.
Rack & site	A physical rack layout and site map: place each miner in its slot so alerts and searches point you at real hardware, not just a name.
Pool speed test	Measures stratum latency from the phone's current network to your configured pools — handy for comparing pools or diagnosing a slow link before blaming a miner.
Live Bitcoin	Opens the Hi3 live Bitcoin network dashboard (blocks, difficulty, mempool at a glance).

Tip: the per-miner tuning optimizer (efficiency curve across saved auto-tune sweeps with a best-efficiency / best-hashrate recommendation) lives on each miner's detail screen alongside the autotuner.

10 · Backups, exports & your data



Update Hashkit, Data & Exports and Demo mode in Settings

- **Fleet telemetry CSV** — the last 7 days of readings for spreadsheets.
- **Backup miners & schedules** — a JSON backup of miners, groups, schedules, notes and settings. Protect it with a passphrase (AES-256) if you'll store or share it; restore merges safely onto another device.
- **Diagnostics bundle** — for bug reports; wallet addresses and IPs are redacted.
- **Retention.** Raw telemetry is downsampled to hourly aggregates after a few days, so charts stay fast and storage stays small even after months.

Back up your data: everything lives on the device. If you switch phones, take a backup first — an encrypted backup restores everything including maintenance notes and photos.

11 · Updating the app

Sideload builds update themselves from the official GitHub releases — no store involved:

- Open **Settings** → **Update Hashkit** → **Check for updates**. The app compares your installed version with the latest signed release.

- If an update exists you'll see **Download & install vX.Y.Z** with its size. Tap it: the APK downloads with a progress bar and installs over the top — **miners, history, notes and settings are all kept** (same signing key, same package).
- The first time, Android may ask you to allow "install unknown apps" for Hashkit — the app opens the right settings screen for you. The first update also shows one system confirmation tap; after that, updates install fully automatically.

Note: updates are checked only when you tap the button — the app never phones home on its own. The optional firmware-update check (section 8) is a separate, opt-in setting.

12 · Troubleshooting & FAQ

Symptom	What to check
Miner shows offline but its web UI works	Same network/VLAN? Tailscale up? Try the miner's IP in a browser on the phone. Pull down to refresh, or raise the poll interval if the miner's web server is slow.
Scan finds nothing	Scans cover private ranges up to /22 only. Check the CIDR matches your subnet; add extra subnets in Settings; try mDNS for AxeOS devices.
"You're on the latest version" but you expected an update	The check compares against the newest <i>published GitHub release</i> . If a release hasn't been published yet, the app is correct.
No temperature / controls on an Antminer-class device	That firmware family is monitoring-only (hashrate, shares, uptime, pool) until its control API is verified. This is deliberate — no guessed commands.
Alerts arrive late with the app closed	Background checks run every 15+ minutes and Android may defer them. Exclude Hashkit from battery optimization for the most reliable delivery.
Efficiency shows "calc"	The firmware doesn't report wall power; the value is calculated from reported figures and marked accordingly.